

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2024

Subject: 031001111 Mechanics I

Section: 15-18

Date: 5 August 2024

Time: 13.00-15.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 6 pages (including this page) and they are worth 35 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

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Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

|       |                                                                                                                                                                                                                                                                        | Part I | Part II |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| CLO 1 | Describe the principle and concept of measurement, dimension and significant figure. Report the results of measurements of physical quantities accurately and appropriately.                                                                                           | 1,4-5  | 2-3     |
| CLO 2 | Calculate of measurement discrepancies into account when presenting results. Show the experimental results in graph form. Able to analyze and interpret meaning from straight line graphs                                                                              |        | 1,5,7   |
| CLO 3 | Describe and calculation of vector and scalar physics quantities                                                                                                                                                                                                       |        | 6       |
| CLO 4 | Explain and calculate quantify and find the relationship between the position, displacement, velocity and acceleration of an object's motion in time. with constant acceleration from graphs and equations Including vertical movement under the gravity of the Earth. | 2-3    | 4,8-9   |

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Asst.Prof. Dr. Dusit Ngamrunroj

Dr. Pinthudit Klinkajorn

Examiner

**Part I** Multiple choices.

Choose the best answer for each question (1 point each)

1. The mass of an object is  $2.4 \times 10^4$  kg. What is the mass expressed in megagrams?  
 a.  $2.4 \times 10^1$  Mg      b.  $2.4 \times 10^{-2}$  Mg      c.  $2.4 \times 10^{-3}$  Mg      d.  $2.4 \times 10^{13}$  Mg
  
2. If a car is traveling at 90 km/h. What is the speed in m/s?  
 a. 0 m/s      b. 25 m/s      c. 90 m/s      d. 324 m/s
  
3. A man drives a boat is 88.00 km in 120.00 mins. What is the speed of boat in km/h?  
 a. 0.73 km/h      b. 12.22 km/h      c. 44.00 km/h      d. 85.00 km/h
  
4. A student uses a measuring device with a resolution of 0.01 cm. to measure the diameter of a coin is 2.59 cm. When considering significant figures. He should record the value of the cross-sectional area according to which of the following?  
 a. 5.27065 cm      b. 5.2707 cm      c. 5,271 cm      d. 5.27 cm
  
5. Which of the following quantities does not have the same number of significant figures as the other quantities?  
 a. 0.0200      b. 123      c. 1.24      d. 0.05

|   | a | b | c | d |
|---|---|---|---|---|
| 1 |   |   |   |   |
| 2 |   |   |   |   |
| 3 |   |   |   |   |
| 4 |   |   |   |   |
| 5 |   |   |   |   |

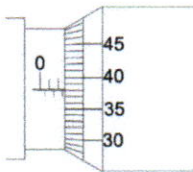
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**Part II Writing**

1. A particle moves in a straight line from rest with an acceleration of  $a$  for time  $t_1$ . It continues with constant velocity for time  $t_2$ , after which it slows down with a continuous deceleration equal to half the magnitude of the initial acceleration until rest. What is the average velocity of the particle during all this movement? (5 points)

2. If  $v$  and  $u$  is velocity,  $a$  is acceleration and  $t$  is time, what is unit of constant  $k$  in the relationship  $v = u + kat^2$ ? (3 points)

3. What is the reading value from the micrometer below? (2 points)



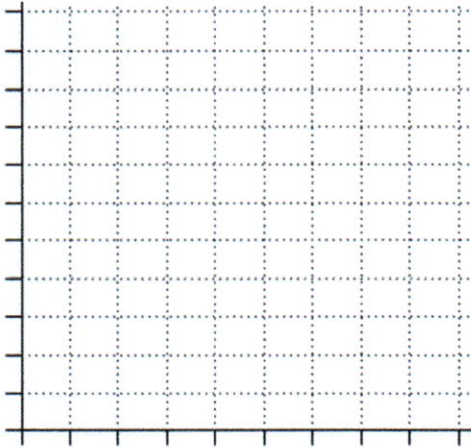
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4. A man leaves his house on a motorcycle to go to his workplace 0.6 km away. If he wants to get there in 2 mins. How much the accelerate of motorcycle in  $m/s^2$ ? (3 points)

5. From the study of the relationship between force and acceleration with the following information:

|                                |     |     |     |     |
|--------------------------------|-----|-----|-----|-----|
| Force (N)                      | 1   | 2   | 3   | 4   |
| acceleration( $\text{m/s}^2$ ) | 0.1 | 0.2 | 0.6 | 0.8 |

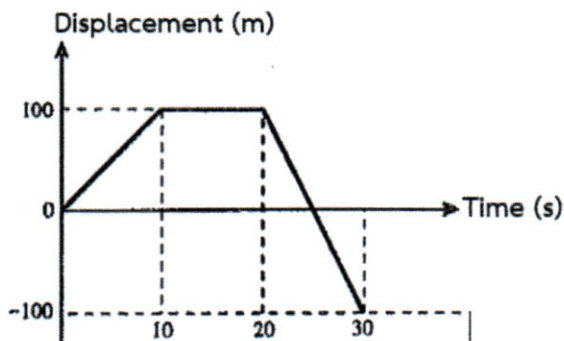
Determine to analyze the relationship of force (F) and acceleration (a) in terms of linear equations. Predict the acceleration when force  $F = 3.5 \text{ N}$ ? (3 points)



6. A car travels at a constant speed of  $36 \text{ km/h}$  for 1 minute and then the engine stops. Make the car continue running with a constant delay. The time it takes from when the engine turns off to when the car comes to rest is 30 seconds. Find the total distance traveled by the car? (3 points)

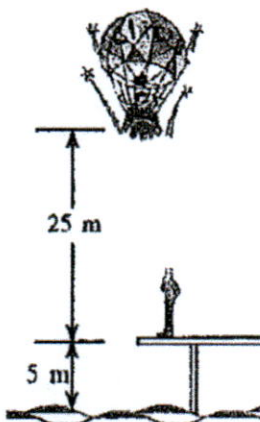
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7. From the graph between linear displacement and time as shown below. Find average velocity during 0 s up to 25 s? (3 points)



8. A ball is thrown directly downward with an initial speed of 10 m/s from a height of 40 m. What is its speed when it strikes the ground? (3 points)

9. The bridge is 5 meters above the surface of the water. A balloon is moving towards the ground and while the balloon is 25 m above the bridge, the person on the balloon and the person standing on the bridge release a rock at the same time. It turned out that both rocks fell to the surface of the water at the same time. What is the speed of the balloon falling from the bridge level in m/s? (5 points)



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